

EXAM 7 – SPRING 2012

24. (2 points)

a. (0.5 point)

Contrast a capital model calibrated to rating agency capital with one calibrated to an economic capital level.

b. (0.5 point)

Explain how an internal model calibrated to an economic impairment earnings level is different than the models described in part a. above.

c. (1 point)

Describe two examples of why it may not be practical to use a single internal model for multiple ERM, operational, and capital applications.

END OF EXAMINATION

Question 24 Sample Answer

Solution 1

- a) Rating agency capital is the amount of capital needed to maintain a certain rating. It can be calculated based on the rating agency's factor based model or adjusted internal model of the insurer.

Economic capital is the amount of capital to keep the probability of ruin of an insurer below a target level. Usually based on an internal model of the insurer and at different confidence level.

- b) Economic impairment earning usually use a much lower confidence level than the 2 above.
- c)
- 1) A single firm wide internal model is difficult to incorporate day-to-day management decisions.
 - 2) A single model may be simpler than needed for product pricing etc.

Solution 2

- a) Rating agency capital is one capital required by the rating agency to maintain credit rating. This rating agency model.

Economic capital level is the capital required for the company to maintain its solvency. It reflects company's own risk measures.

- b) Its internal model estimated at a lower return period, i.e. lower confidence level. It's not capital $\Rightarrow$ it measures losses that will put company in economic impairment.
- c)
- 1) Models for business units may be more complex than the single internal model, e.g. pricing, bonus structure calculation is complex.
 - 2) A single internal model may not be able to handle day-to-day business decision making. e.g., a simple decision of whether underwriting a policy may take a day to run, so it's not feasible due to long run time.

Examiner Comment

It was common for the candidates to know the exact definitions of rating agency capital and economic capital. Depending on how complete the definitions were an explicit "contrast" of the two types was not always needed since it was inherent in the full definitions. Some candidates

contrasted by assuming one produces a higher capital level than the other. However in definition and in practice this can vary by company.

For the b. part, it was common for the candidates to know the exact definition of economic impairment earnings level and successfully compare to part a. Occasionally, candidates would interchange the definitions of economic impairment with economic capital.

For the c. part the question was answered correctly the majority of the time. A few candidates provided only one example. We did want to see some commonality in the answer with the examples found in the syllabus article. In responses that were too generic or simply irrelevant deductions were made.